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Pharmacokinetics of Intravenous Paracetamol in Elderly Patients

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Abstract

Background and Objectives: Intravenous paracetamol (*N*-acetyl-paraminophenol, acetaminophen) is a widely used nonopioid analgesic which has become popular in the treatment of pain in many patient groups, including the elderly. Although intravenous paracetamol has been studied widely in clinical analgesia studies, there is little information on its pharmacokinetics in the elderly. We designed this study to determine the pharmacokinetics of intravenous paracetamol in very old patients and to compare them with that of younger patients. We also considered the effect of adenosine triphosphate-binding cassette G2 protein (*ABCG2*) genotype and renal function on paracetamol pharmacokinetics in these patients.

Methods: We compared the pharmacokinetics of intravenous paracetamol in four groups of ten patients, aged 20–40, 60–70, 70–80 and 80–90 years, undergoing orthopaedic surgery. Paracetamol 1000 mg was given by infusion over 15 minutes. Plasma concentrations of paracetamol and its glucuronide and sulphate conjugates were measured for 24 hours with a high-performance liquid chromatographic method and *ABCG2* genotype was determined. Glomerular filtration rate (GFR) was estimated from age, sex and serum creatinine of the patient.

Results: In the group aged 80–90 years, the mean value of the area under the plasma concentration-time curve extrapolated to infinity (AUC_∞) of paracetamol was 54–68% higher than in the two youngest groups. Paracetamol clearance showed a statistically significant dependence on age group, whereas volume of distribution during elimination and elimination half-life were associated with age group and sex, respectively. Based on mean AUC_∞ of paracetamol glucuronide and paracetamol sulphate, the oldest patients had 1.3- to 1.5-fold greater exposure to these metabolites than patients aged 20–40 years. *ABCG2* genotype did not affect paracetamol pharmacokinetics. There was a linear correlation between the values of AUC_∞ of paracetamol, its glucuronide and sulphate metabolites and GFR.

Conclusion: Age and sex are important factors affecting the pharmacokinetics of paracetamol. The higher the age of the patient, the higher is the exposure to paracetamol. Female sex is associated with increased paracetamol concentrations but *ABCG2* genotype does not seem to affect paracetamol pharmacokinetics.

Trial registration number (EudraCT): 2006-001917-14

Background

Paracetamol (*N*-acetyl-paraminophenol, acetaminophen) is a widely used nonopioid analgesic which lacks many of the adverse effects that the classic nonsteroidal anti-inflammatory drugs have.[1] Multimodal analgesia for acute postoperative pain with opioids in conjunction with paracetamol has become very popular as intravenous paracetamol became commercially available in 2002. Although intravenous paracetamol has been studied widely in clinical analgesia studies,[2-4] there is little information on its pharmacokinetics in the elderly. The currently available pharmacokinetic data are mainly based on studies performed in patients under 70 years of age or in healthy subjects. Nevertheless, a growing number of patients requiring analgesics are much older, up to 90 or even 100 years of age. Challenges of drug therapy in the elderly have been the subject of several reviews and other articles in medical literature in recent years.[5-8] We designed this study to determine the pharmacokinetics of intravenous paracetamol in very old patients and to compare them with that of younger patients. Because the relative amounts of the major metabolites of paracetamol in children are dependent on the age of the patient, we measured the concentrations of both the (parent) paracetamol and its glucuronide and sulphate metabolites.[9] We also considered the effect of adenosine triphosphate-binding cassette G2 protein (*ABCG2*) genotype and renal function on paracetamol pharmacokinetics in these patients. ABCG2 protein is an efflux transporter that is expressed in various normal tissues such as small intestine, colon, liver, kidney, stem cells, placenta, mammary gland, brain and heart, as well as in cancer cells.[10] It mediates the cellular efflux of a wide variety of xenobiotics including sulphate and glucuronide metabolites of some compounds such as paracetamol.[11]

Patients and Methods

Patients

After obtaining institutional approval and informed written consent, 40 patients aged 20–90 years whose physical status was classified as I–IV according to the American Society of Anaesthesiologists' physical status classification were included in this study. The study protocol was approved by the Finnish Medicines Agency and was thereby reported to the European EudraCT clinical trials register and was given the code number 2006-001917-14. Ten patients were recruited to each of the following four groups: patients aged 20–40, 60–70, 70–80 and 80–90 years. Eight patients were calculated to be required in each group to demonstrate a 30% difference in the largest and smallest mean values of the paracetamol area under plasma concentration-time curve (AUC) extrapolated to infinity (AUC_∞) at a level of significance of $p < 0.05$ and power of 80%. The youngest patients were scheduled for arthroscopic anterior cruciate ligament operation of the knee and the older patients for elective knee prosthesis operation. Patients using strong inhibitors or inducers of cytochrome P450 enzymes or having significant hepatic, renal, neurological, haematological, endocrine, metabolic or gastrointestinal diseases or a body mass index >35 kg/m^2 were excluded from the study. Patients were judged to have a significant hepatic, renal, neurological, haematological or gastrointestinal disease if they had a clinical diagnosis of any of the above diseases. Patients with diabetes mellitus were eligible unless they had significant renal involvement. Renal function was evaluated by estimating the glomerular filtration rate (GFR) from age and sex of the patient and his/her serum creatinine (SCr) by using abbreviated Modification of Diet in Renal Disease study equation (equation 1):[12]

$$GFR = 186 \times 88.4^{1.154} \times SCr^{-1.154} \times Age^{-0.203} \times (0.742 \text{ if female}) \quad \text{(Eq. 1)}$$

Anaesthesia and Treatment of Postoperative Pain

All patients were anaesthetized with an intrathecal injection of 5 mg/mL plain bupivacaine. An epidural catheter was placed for postoperative pain management. The epidural solution contained 1.0 mg/mL plain bupivacaine and 5 μg/mL fentanyl. The infusion pump was set to deliver 1–7 mL/h of the solution to the epidural catheter. In case of insufficient pain relief, as determined by subjective visual analogue scale (VAS) score of 3 (minimum 0–maximum 10), patients were set to receive morphine 3 mg intravenously at 15-minute intervals until VAS score was under 4.

Postoperative Care

All patients stayed at the post-anaesthesia care unit (PACU) until the first postoperative morning. Standard laboratory tests including complete blood count, C-reactive protein, sodium and potassium were taken in the PACU in the first postoperative morning from all patients undergoing knee prosthesis operation and from other patients if clinically indicated. After knee prosthesis surgery the patients were hospitalized for 4–5 days. After anterior cruciate ligament operation the patients were discharged within 24 hours after the operation. During their hospital stay all patients were observed according to the normal routine of our hospital. Further laboratory tests were taken only when clinically indicated.

Administration of the Study Drug and Blood Sampling

The study medication, 1000 mg of intravenous paracetamol (Perfalgan 10 mg/mL 100 mL solution; Bristol-Myers Squibb, Agen, France), was administered using an automated infusion device (Braun Infusomat Space Infusor; B. Braun Melsungen AG, Melsungen, Germany) over 15 minutes. Timed venous blood samples were drawn into heparinized tubes before the administration of paracetamol and at 7.5 and 15 minutes during the infusion. After completion of the infusion, samples were drawn at 2.5, 5, 10, 15, 20, 30, 45 and 60 minutes and at 1.5, 2, 3, 4, 5, 6, 8, 10, 12, 18 and 24 hours.

Bioanalytical Assay

Plasma concentrations of paracetamol and its major metabolites, paracetamol glucuronide and paracetamol sulphate, were determined using a minor modification of a high-performance liquid chromatographic method.[13] We precipitated plasma proteins using methanol, instead of perchloric acid, to avoid a hydrolysis of paracetamol sulphate in the acidic medium. The lower limits of quantification were 0.25 mg/L for paracetamol and paracetamol sulphate and 0.5 mg/L for paracetamol glucuronide. The interday coefficient of variation (CV) for paracetamol was 12.8%, 12.5% and 5.1% at 0.398, 2.01 and 10.1 mg/L, respectively (n = 38). The CV for the glucuronide metabolite was 11.5% and 4.5% at 1.76 and 9.48 mg/L (n = 41), respectively. The corresponding values for paracetamol sulphate were 13.6%, 11.6% and 7.4% at 0.342, 1.72 and 8.74 mg/L (n = 38).

Genotyping for *ABCG2*

Genomic DNA was extracted from whole-blood samples using standard methods (QIAamp DNA Blood Mini Kit; Qiagen, Hilden, Germany). The subjects were genotyped for the *ABCG2* c.421C>A single nucleotide polymorphism (p.Gln141Lys; rs2231142) by allelic discrimination with a TaqMan Drug Metabolism Genotyping Assay (assay ID: C__15854163_70; Applied Biosystems, Foster City, CA, USA).[14]

Pharmacokinetic Measurements

Maximum plasma concentration (C_{max}) and time to reach C_{max} (t_{max}) of paracetamol and its metabolites were observed directly from the data. The terminal elimination rate constant (λ_z) was determined by regression analysis of the log-linear part of the curve. The elimination half-life ($t_{½}$) was calculated from $t_{½} = \ln 2/\lambda_z$. The AUC_∞ for paracetamol and the metabolites and the total plasma clearance (CL) and volume of distribution during elimination (V_z) for paracetamol were calculated using standard noncompartmental methods (WinNonlin pharmacokinetic program, version 4.1; Pharsight Corporation, Mountain View, CA, USA). Because a standard dose of 1000 mg of paracetamol was administered to all patients, the standardized AUC_∞ ($sAUC_\infty$) for paracetamol and the metabolites was calculated by dividing the AUC_∞ of paracetamol by paracetamol dose expressed in mg per kg bodyweight. The metabolite-to-parent drug AUC ratios (AUC_m/AUC_p) were calculated so as to afford a comparison of the relative abundance of each metabolite.

Statistical Analysis

Because all 40 patients had either *ABCG2* c.421CA or c.421CC genotype, the possible effect of *ABCG2* genotype on pharmacokinetic variables was tested using Student's t-test prior to other statistical analyses. Thereafter, all pharmacokinetic

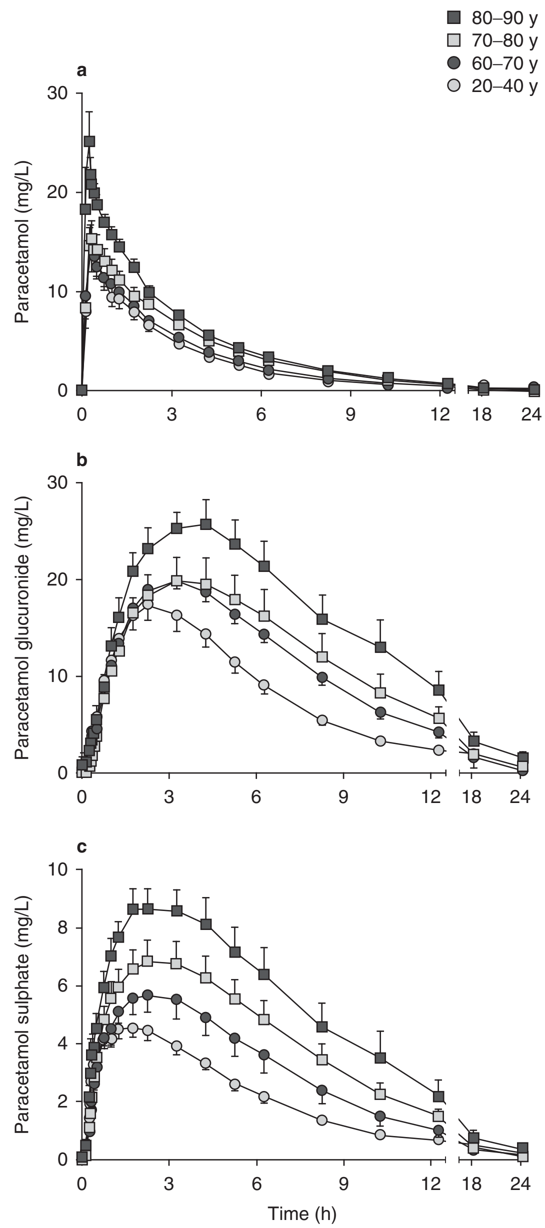

Fig. 1. Mean ± standard error of the mean (SEM) plasma concentrations of (**a**) paracetamol, (**b**) paracetamol glucuronide and (**c**) paracetamol sulphate after a single intravenous dose of paracetamol 1000 mg in 40 patients aged 20–90 years (10 patients in each group) who were undergoing orthopaedic surgery.

variables except t_{max} were compared using stepwise two-way (age group, sex, age group × sex) analysis of variance. The values for t_{max} were compared using the Kruskal-Wallis test followed by the Mann-Whitney U-test. $p < 0.05$ was considered as statistically significant. The results are expressed as mean ± standard deviation (SD), except in figure 1 where, for clarity, mean ± standard error of the mean (SEM) values are given. All data were analysed using the statistical program SYSTAT for Windows (version 10.2; Systat Software, Richmond, CA, USA).

Results

Patient Characteristics

Patient characteristics are shown in table I. No clinical signs of paracetamol toxicity were observed in any of the patients. Based on the availability of serum creatinine determinations, GFR could be estimated for all patients who were older than 60 years but only for three patients in the group aged 20–40 years. The youngest and oldest groups differed significantly from the others in many respects. On the basis of the American Society of Anaesthesiologists' physical status classification the youngest group was healthier than the others. In the oldest group there was an overrepresentation of females as compared to the other groups. As a result, the oldest group also weighed less and was shorter than the others. All patients were Caucasian. All except two male patients in the youngest group were non-smoking.

Effect of Age on Paracetamol Pharmacokinetics

In the group aged 80–90 years the mean value of AUC_{∞} of paracetamol obtained after the dose of 1000 mg of paracetamol was 54–68% higher than in the two youngest groups, respectively (figures 1 and 2; see also table S-1 in Supplemental Digital Content 1; http://links.adisonline.com/CPZ/A13). The $sAUC_{\infty}$ was higher in the two oldest groups as compared to the two youngest groups and there was also an association with sex. The calculated pharmacokinetic variables with the results of the statistical analysis are shown in table II. CL showed a statistically significant dependence on age group, whereas V_z and $t_{½}$ were associated with age group, sex and age group × sex interaction. C_{max} was only associated with sex.

Based on the mean values of AUC_{∞} of paracetamol glucuronide and paracetamol sulphate, the oldest patients had 1.3- to 1.5-fold greater exposure to these metabolites than patients aged 20–40 years. The individual values of AUC_m/AUC_p ratios suggested that the relative abundance of paracetamol glucuronide was dependent on patient age or sex as the mean AUC_m/AUC_p ratio was smaller in young adults than the others.

Effect of Renal Function and *ABCG2* Genotype on Paracetamol Pharmacokinetics

There was a linear correlation between the values of AUC_{∞} of paracetamol, its glucuronide and sulphate metabolites and the estimated GFR (figure 3). The values for squared correlation coefficient (r^2) varied from 0.19 to 0.63 and they were clearly higher for the association between metabolite AUC_{∞} and estimated GFR.

The relationship of *ABCG2* c.421C>A genotype and the AUC_{∞} values for paracetamol and its glucuronide and sulphate metabolites were measured but no statistically significant differences were observed.

Analgesia

None of the patients required any additional pain relief to paracetamol and epidural infusion of bupivacaine and fentanyl. Naloxone was given to eight patients (two aged 20–40 years,

Table I. Patient characteristics

Age group (y)	Sex (n, male/female)	ASA physical status (n, I/II/III/IV)	*ABCG2* genotype (n, c.421CC/c.421CA/c.421AA)	Age (y)[a]	Weight (kg)[a]	Height (cm)[a]	BMI (kg/m²)[a]	Estimated GFR (mL/min/1.73 m²)[a]
20–40	7/3	8/2/0/0	8/2/0	27 ± 5	81 ± 14	177 ± 10	25.7 ± 2.2	108 ± 10
60–70	5/5	0/6/4/0	7/3/0	66 ± 3	83 ± 9	169 ± 9	29.0 ± 1.5***	82 ± 16
70–80	6/4	0/5/5/0	9/1/0	77 ± 3	82 ± 13	169 ± 10	28.9 ± 3.3***	76 ± 17***
80–90	1/9	0/6/2/2	9/1/0	84 ± 3	68 ± 8*	160 ± 7**	26.3 ± 2.2	64 ± 19**
p-Value[b]	0.040	<0.0001	0.592		0.012	0.003	0.004	0.003

a Values are expressed as mean ± SD unless specified otherwise.

b p-Values determined by analysis of variance, except for sex, ASA physical status and *ABCG2* genotype, which were determined by chi-square test.

ASA = American Society of Anaesthesiologists; **BMI** = body mass index; **GFR** = glomerular filtration rate; **SD** = standard deviation; * $p < 0.05$ vs age groups 20–40, 60–70 and 70–80 y, ** $p < 0.01$ vs age group 20–40 y, *** $p < 0.05$ vs age group 20–40 y.

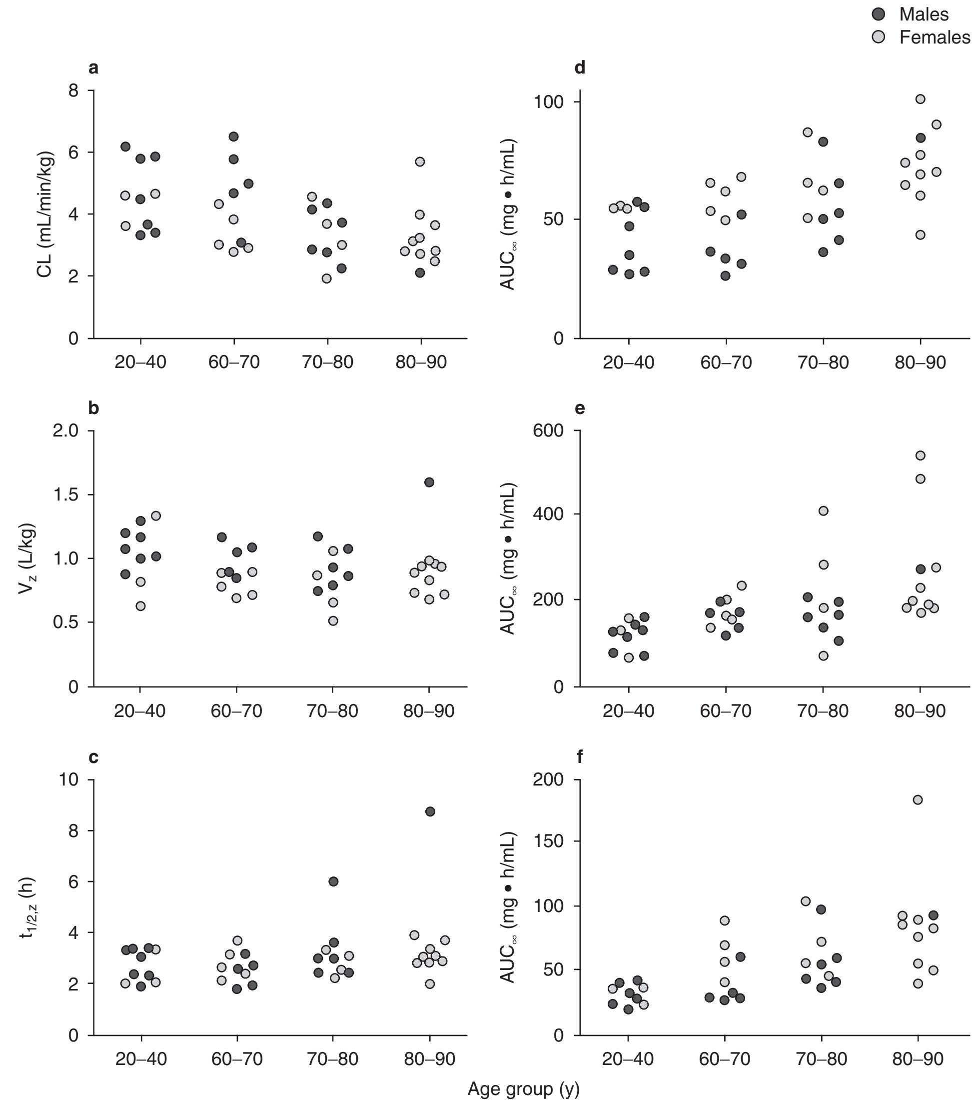

Fig. 2. Individual values for (**a**) total plasma clearance (CL), (**b**) volume of distribution during elimination (V_z) and (**c**) terminal elimination half-life ($t_{½,z}$) after a single intravenous dose of paracetamol 1000 mg in 40 patients aged 20–90 years (10 patients in each group) who were undergoing orthopaedic surgery; and values of area under the plasma concentration-time curve extrapolated to infinity (AUC_{∞}) for (**d**) paracetamol, (**e**) paracetamol glucuronide and (**f**) paracetamol sulphate.

two aged 60–70 years, one aged 70–80 years and three aged 80–90 years) for itching.

Discussion

The administration of a standard dose of 1000 mg of intravenous paracetamol resulted in 36–68% higher exposure to paracetamol in patients aged 70–80 and 80–90 years than in patients aged 20–40 years. The higher concentrations were caused by lower CL and smaller V_z in the oldest patients. These age-related changes in the primary pharmacokinetic variables were also reflected in the observed values for C_{max} and $t_{½}$ which were increased in the oldest group. Patients aged 80–90 years had, on the average, 90% higher paracetamol plasma concentrations than the young adults 8 hours after its infusion (see table S-1 in Supplemental Digital Content 1).

Previous studies have given inconclusive information on the pharmacokinetics of paracetamol in the elderly. Some of the discrepancies might have been caused by short and sparse sampling of blood following the administration of paracetamol. In many studies paracetamol has been given orally which has not allowed the determination of the primary pharmacokinetic parameters CL and V_z. Triggs et al.[15] and Miners et al.[16] studied the oral pharmacokinetics of paracetamol in young and elderly subjects. Both study groups concluded that pharmacokinetics of paracetamol were essentially similar in the two age groups. Both studies, however, suffered from a short sampling period of only 6–8 hours. Briant et al.[17] reported

Table II. Estimates[a] of noncompartmental pharmacokinetic parameters for paracetamol and its primary metabolites in patients undergoing orthopaedic surgery. Ten patients in each group

Age group (y)	CL (mL/min/kg)	V_z (L/kg)	t_{max} (h)	C_{max} (mg/L)	$t_{1/2}$ (h)	AUC_∞ (mg • h/mL)	$sAUC_\infty$ (kg • h/mL)	AUC_m/AUC_p
Paracetamol								
20–40	4.6 ± 1.1	1.04 ± 0.22	0.31 ± 0.07	16.3 ± 4.5	2.7 ± 0.6	44 ± 13	3.5 ± 0.8	
60–70	4.2 ± 1.3	0.90 ± 0.16	0.32 ± 0.08	16.6 ± 9.0	2.6 ± 0.6	48 ± 15	3.9 ± 1.1	
70–80	3.3 ± 0.9	0.87 ± 0.20	0.35 ± 0.15	15.9 ± 5.3	3.2 ± 1.1	60 ± 17	4.9 ± 1.5	
80–90	3.3 ± 1.0	0.92 ± 0.26	0.26 ± 0.05	25.8 ± 9.2	3.6 ± 1.9	74 ± 16	5.0 ± 1.2	
p-Value for age group[b]	0.004	0.007	0.325		<0.001	0.008	0.002	
p-Value for sex[b]		<0.001	NA	<0.001	<0.001	0.010		
p-Value for age group × sex[b]	0.061	0.040	NA		<0.001		0.057	
Paracetamol glucuronide								
20–40			2.3 ± 0.4	17.6 ± 5.8	3.1 ± 0.7	117 ± 34	9.5 ± 2.9	2.80 ± 0.95
60–70			3.1 ± 0.7	20.3 ± 2.5	3.4 ± 0.7	168 ± 35	13.9 ± 3.3	3.76 ± 1.07
70–80			3.5 ± 0.6	20.7 ± 8.7	4.0 ± 0.6	192 ± 95	15.9 ± 8.6	3.26 ± 1.21
80–90			3.9 ± 0.7	27.0 ± 7.7	4.3 ± 1.0	272 ± 131	18.3 ± 8.6	3.81 ± 1.87
p-Value for age group[b]			<0.001	0.022	0.009	0.003	0.031	0.291
p-Value for sex[b]			NA					
p-Value for age group × sex[b]			NA					
Paracetamol sulphate								
20–40[c]			1.6 ± 0.5	4.8 ± 0.9	3.1 ± 0.7	31 ± 8	2.5 ± 0.7	0.70 ± 0.18
60–70[c]			2.3 ± 0.6	6.0 ± 1.9	3.7 ± 1.3	48 ± 22	4.0 ± 1.9	1.02 ± 0.22
70–80			2.8 ± 0.8	7.1 ± 2.6	3.6 ± 0.7	61 ± 23	4.9 ± 2.0	1.03 ± 0.34
80–90			2.4 ± 0.9	9.2 ± 2.6	4.0 ± 1.1	84 ± 39	5.6 ± 2.5	1.13 ± 0.39
p-Value for age group[b]			0.009	0.012	0.259	<0.001	0.007	0.025
p-Value for sex[b]			NA	0.043				
p-Value for age group × sex[b]			NA					

a Estimates are expressed as mean ± SD.

b p-Values determined by stepwise two-way analysis of variance, except for t_{max}, which was determined by Kruskal-Wallis test.

c n = 9, since paracetamol sulphate concentrations could not be quantified because of disturbing peaks.

AUC = area under plasma concentration-time curve; **AUC_m/AUC_p** = metabolite-to-parent drug AUC ratio; **AUC_∞** = AUC extrapolated to infinity; **CL** = total plasma clearance; **C_{max}** = maximum plasma concentration; **NA** = not applicable; **$sAUC_\infty$** = standardized AUC_∞ calculated by dividing the AUC_∞ of paracetamol by paracetamol dose expressed in mg per kg bodyweight; **SD** = standard deviation; **$t_{1/2}$** = elimination half-life; **t_{max}** = time to reach C_{max}; **V_z** = volume of distribution during elimination.

a prolongation of paracetamol $t_{1/2}$ as a result of the reduction of CL, but the calculations were based on only four samples per patient which makes the reliable assessment of pharmacokinetics difficult. Divoll et al.,[18] on the other hand, observed that higher age was associated with lower CL and smaller V_z but unchanged $t_{1/2}$ of paracetamol. They also reported clear-cut pharmacokinetic differences between the sexes because the values for CL and V_z were lower in women than in men.

In addition to the methodological differences between previous paracetamol studies in the elderly and our study, the observed differences might have been due to the heterogeneity of the elderly study population. In some of the above studies the elderly patients have been healthy while in others they might have suffered from disabling health conditions.[15-18] Indeed, it has been suggested that it is not the age itself which is responsible for the majority of the changes in paracetamol pharmacokinetics but it is the general health condition which explains the differences. Wynne et al.[19] studied the association of age and frailty with paracetamol metabolism in man. They concluded that the physical state of the elderly person is strongly associated with the capacity of the liver to eliminate paracetamol from the body because paracetamol pharmacokinetics in healthy elderly people were in fact similar to that in young adults.

In the present study the mean age of the oldest group was 84 years and the oldest patient was almost 89 years old. There was an overrepresentation of females in the oldest group which somewhat complicated the analysis of age on the pharmacokinetics of paracetamol. However, when age and sex are taken into account in the stepwise analysis of variance, the primary pharmacokinetic parameters CL and V_z were dependent on both of these factors. The values for CL and V_z expressed per kg of bodyweight are lower in women than in men and also lower in the elderly as compared to youngest adult patients in this study. All our patients were living at home before surgery and therefore none of them can be regarded as physically frail or having severe disabling conditions. Severe disabling conditions would most likely have had an additional effect on paracetamol pharmacokinetics.[19]

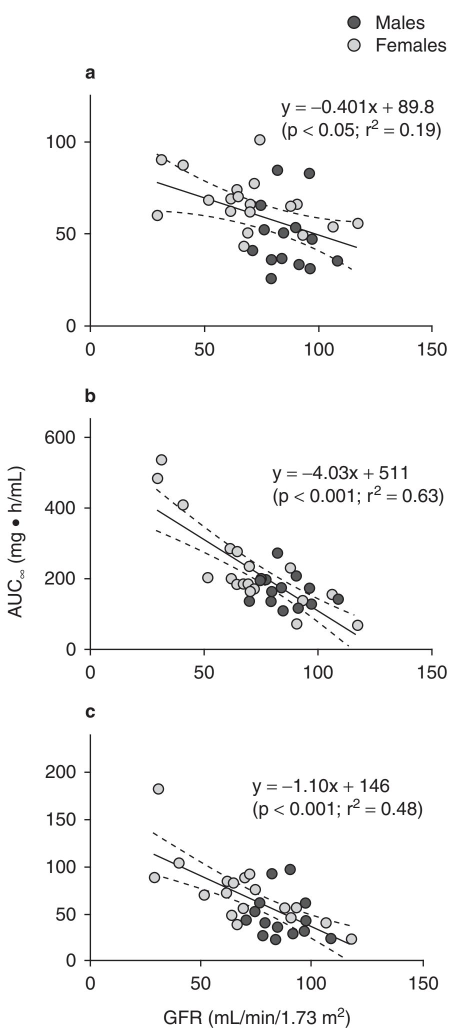

Fig. 3. Relationship between area under the plasma concentration-time curve extrapolated to infinity (AUC_∞) and glomerular filtration rate (GFR) for (**a**) paracetamol, (**b**) paracetamol glucuronide and (**c**) paracetamol sulphate after a single intravenous dose of paracetamol 1000 mg in 40 patients aged 20–90 years who were undergoing orthopaedic surgery. The dashed lines represent the 95% confidence intervals for each regression line (solid line).

ABCG2 protein (also known as breast cancer resistance protein) is an efflux transporter that is expressed in various normal tissues such as small intestine, colon, liver, kidney, stem cells, placenta, mammary gland, brain and heart, as well as in cancer cells.[10] It mediates the cellular efflux of a wide variety of xenobiotics including sulphate and glucuronide metabolites of some compounds such as paracetamol.[20] *ABCG2* genotype could therefore have an effect on the elimination of paracetamol and especially its metabolites from the body. Previous studies have demonstrated that the *ABCG2* c.421A allele is associated with reduced transport activity of ABCG2, both *in vitro* and *in vivo*.[11,14] However, in our study with no patients with the c.421AA genotype and seven patients with the c.421CA genotype, *ABCG2* genotype had no quantifiable effect on paracetamol pharmacokinetics. Further studies are therefore warranted to shed light on the possible effect of *ABCG2* genotype on paracetamol pharmacokinetics.

The values for AUC_∞ of paracetamol glucuronide and sulphate conjugates increased with age as did C_{max} which can be explained by changes in GFR.[21] Plasma concentrations of paracetamol and its glucuronide and sulphate conjugates are increased in patients with renal failure.[21] Although our study group did not include any patients with manifest renal failure, there were patients whose estimated GFR implicated mild to moderate renal dysfunction. However, renal dysfunction has only a minor effect on the parent paracetamol.[21] This is in good agreement with the present study where the slope of the regression equation describing the relationship between the GFR and AUC_∞ of paracetamol had a value of 0.401. It can be calculated, using the regression equation obtained for paracetamol (figure 3), that the deterioration of GFR from the value of 90 mL/min/1.73 m^2 (lower limit of the normal range) to the smallest values observed in the present study would increase the exposure to paracetamol only by approximately 40% which is unlikely to necessitate dose adjustments during short-term administration of paracetamol. Our results are in line with previous studies[22] and it can therefore be concluded that the current prescribing information concerning the dosing of paracetamol appears to be appropriate and dose adjustments are not needed for patients having GFR higher than 50 mL/min/1.73 m^2.

The current practice is to use a standard dose for all adults irrespective of patient age.[23] Although paracetamol is a safe drug when used according to dosage recommendations it is a drug with relatively narrow therapeutic index.[24] Indeed, the safety of commonly recommended doses of paracetamol has recently been questioned.[25] Studies have shown that sensitive individuals may display disturbances in hepatocellular integrity even after a single dose of paracetamol.[26] If a standard dose of 1000 mg of intravenous paracetamol is administered to all adults irrespective of their age and sex, we can assume that, on the average, patients aged 80–90 years will have a 70% larger exposure to paracetamol than young adults. In the present study the highest value of AUC_{∞} of paracetamol in the group aged 80–90 years was 3.8-fold as compared to the corresponding smallest value of AUC_{∞} in young adults (figure 2). If we had standardized the dose per bodyweight, the oldest group would have had a 40% larger exposure to paracetamol, on the average, and the highest difference in AUC_{∞} values would be 3.3-fold. Thus, the use of standard doses per bodyweight would not protect against paracetamol-induced hepatocellular toxicity as compared to the current practice.

Although paracetamol is mainly metabolized to glucuronide and sulphate metabolites, a minor hydroxylated metabolite (*N*-acetyl-*p*-benzoquinoneimine [NAPQI]) is produced in small amounts by cytochrome P450 isoenzymes in the liver and kidney.[27,28] The amount of NAPQI produced after normal doses of paracetamol is usually completely detoxified by conjugation with glutathione and excreted as mercaptopurine and cysteine conjugates. In paracetamol overdosage and sensitive individuals, tissue stores of glutathione may become depleted, allowing NAPQI to accumulate and cause liver cell damage.[29] No clinical signs of liver toxicity were observed in our patients after the single intravenous dose of 1000 mg of paracetamol. However, our study can be criticized for not assessing the liver function routinely with liver function tests. Thus, minor changes in liver function in our patients cannot be excluded.

Conclusion

The clinical implication of this study is that age and sex are important factors affecting the pharmacokinetics of paracetamol. The higher the age of the patient, the higher is the exposure to paracetamol. Female sex is associated with increased paracetamol concentrations but *ABCG2* genotype does not seem to affect paracetamol pharmacokinetics.

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References

1. Hyllested M, Jones S, Pedersen JL, et al. Comparative effect of paracetamol, NSAIDs or their combination in postoperative pain management: a qualitative review. Br J Anaesth 2002; 88: 199-214
2. Elia N, Lysakowski C, Tramér MR. Does multimodal analgesia with acetaminophen, nonsteroidal antiinflammatory drugs, or selective cyclooxygenase-2 inhibitors and patient-controlled analgesia morphine offer advantages over morphine alone? Anesthesiology 2005; 103: 1296-304
3. Remy C, Marret E, Bonnet F. Effects of acetaminophen on morphine side-effects and consumption after major surgery: meta-analysis of randomized controlled trials. Br J Anaesth 2005; 94: 505-13
4. Sinatra RS, Jahr JS, Reynolds LW, et al. Efficacy and safety of single and repeated administration of 1 gram intravenous acetaminophen injection (paracetamol) of pain management after major orthopedic surgery. Anesthesiology 2005; 102: 822-31
5. Hammerlein A, Derendorf H, Lowenthal DT. Pharmacokinetic and pharmacodynamic changes in the elderly: clinical implications. Clin Pharmacokinet 1998; 35: 49-64
6. Butler JM, Begg EJ. Free drug metabolic clearance in elderly people. Clin Pharmacokinet 2008; 47: 297-321
7. White M, Kenny GN, Schraag S. Use of target controlled infusion to derive age and gender covariates for propofol clearance. Clin Pharmacokinet 2008; 47: 119-27
8. Klotz U. Pharmacokinetics and drug metabolism in the elderly. Drug Metab Rev 2009; 41: 67-76
9. Miller RP, Roberts RJ, Fisher LJ. Acetaminophen elimination kinetics in neonates, children, and adults. Clin Pharmacol Ther 1976; 19: 284-94
10. Hardwick LJ, Velamakanni S, van Veen HW. The emerging pharmacotherapeutic significance of the breast cancer resistance protein (ABCG2). Br J Pharmacol 2007; 151: 163-74
11. Kondo C, Suzuki H, Itoda M, et al. Functional analysis of SNPs variants of BCRP/ABCG2. Pharm Res 2004; 21: 1895-903
12. Levey AS, Coresh J, Balk E, et al. National Kidney Foundation practice guidelines for chronic kidney disease: evaluation, classification, and stratification. Ann Intern Med 2003; 139: 137-47
13. Vertzoni MV, Archontaki HA, Galanopoulou P. Development and optimization of a reversed-phase high-performance liquid chromatographic method for the determination of acetaminophen and its major metabolites in rabbit plasma and urine after a toxic dose. J Pharmaceut Biomed 2003; 32: 487-93
14. Keskitalo JE, Zolk O, Fromm MF, et al. ABCG2 polymorphism markedly affects the pharmacokinetics of atorvastatin and rosuvastatin. Clin Pharmacol Ther 2009; 86: 197-203
15. Triggs EJ, Nation RL, Long A, et al. Pharmacokinetics in the elderly. Eur J Clin Pharmacol 1975; 8: 55-62
16. Miners JO, Penhall R, Robinson RA, et al. Comparison of paracetamol metabolism in young adult and elderly males. Eur J Clin Pharmacol 1988; 35: 157-60
17. Briant RH, Dorrington RE, Cleal J, et al. The rate of acetaminophen metabolism in the elderly and the young. J Am Geriatr Soc 1976; 24: 359-61
18. Divoll M, Abernethy DR, Ameer B, et al. Acetaminophen kinetics in the elderly. Clin Pharmacol Ther 1982; 31: 151-6
19. Wynne HA, Cope LH, Herd B, et al. The association of age and frailty with paracetamol conjugation in man. Age Ageing 1990; 19: 419-24
20. Zamek-Gliszczynski MJ, Nezasa K, Tian X, et al. The important role of bcrp (abcg2) in the biliary excretion of sulphate and glucuronide metabolites of acetaminophen, 4-methylumbelliferone, and harmol in mice. Mol Pharmacol 2006; 70: 2127-33
21. Prescott LF, Speirs GC, Critchley JA, et al. Paracetamol disposition and metabolite kinetics in patients with chronic renal failure. Eur J Clin Pharmacol 1989; 36: 291-7

22. Bannwarth B, Pehourcq F, Lagrange F, et al. Single and multiple dose pharmacokinetics of acetaminophen (paracetamol) in polymedicated very old patients with rheumatic pain. J Rheumatol 2001; 28: 182-4
23. Electronic Medicines Compendium UK. Perfalgan 10 mg/mL solution for infusion: summary of product characteristics [online]. Available from URL: http://emc.medicines.org.uk/medicine/14288/SPC/Perfalgan%2010mg/ml%20Solution%20for%20Infusion/ [Accessed 2010 Nov 8]
24. Sweetman S, editor. Martindale: the complete drug reference [electronic version]. London: Pharmaceutical Press, 2006 Sep 1
25. Kaplowitz N. Acetaminophen hepatoxicity: what do we know, what don't we know, and what do we do next? Hepatology 2004; 40: 23-6
26. Watkins PB, Kaplowitz N, Slattery JT, et al. Aminotransferase elevations in healthy adults receiving 4 grams of acetaminophen daily: a randomized controlled trial. JAMA 2006; 296: 87-93
27. Manyike PT, Kharasch ED, Kalhorn TF, et al. Contribution of CYP2E1 and CYP3A to acetaminophen reactive metabolite formation. Clin Pharmacol Ther 2000; 67: 275-82
28. Gelotte CK, Aulier JF, Lynch JM, et al. Disposition of acetaminophen at 4, 6, and 8 g/day for 3 days in healthy young adults. Clin Pharmacol Ther 2007; 81: 840-8
29. American Academy of Pediatrics Committee on Drugs. Acetaminophen toxicity in children. Pediatrics 2001; 108: 1020-4

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